

GIS DATA PIPELINE FOR TUNISIAN ONAGRI DATA - User guide

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OBJECTIVES:

- Normalize inconsistent regional naming across datasets.
- Resolve bilingual (arabic and french) administrative naming conflicts.
- Link statistical datasets to official administrative boundaries.
- Transform non-spatial data into GIS-compatible formats.
- Generate reproducible spatial layers for visualization and analysis (in QGIS for example).

METHODOLOGY:

1- All datasets are dowloaded by hand from the ONAGRI-related data catalog website (<https://catalog.agridata.tn/en/dataset/>) in the folder "Raw data" for each category (Forests, Animal production, Land development, Dams & Irrigated areas, Rainfall, Agricultural products).

2- The R code "1- Data processing & normalization.R" fixes the major inconsistencies between Arabic and French naming conventions in the raw data and normalizes it using a reference dataset "Tunisia.csv" (extracted from GIS files from <https://gadm.org/>).

A unified region_id is added to join tabular data with spatial geometries.

The output datasets are stored automatically in "Clean data".

3- The R code "2- Conversion into GIS data.R" merges the clean data with the Tunisia_regions.shp boundary file using region_id. This attaches geometry to non-spatial data.

4- Each enriched dataset is exported as an individual shapefile stored in the "GIS Layers" folder.

CHALLENGES:

- Lack of standardized region identifiers.
- Most datasets did not include a stable region ID nor coordinates. This forced the painful creation of a synthetic region_id from the reference dataset.
- Mixed-language encoding issues (arabic and french, inconsistent names).
- Absence of spatial data in raw datasets (no XY coordinates)

USER GUIDE:

Note: Ensure the raw datasets and the reference files are in the correct folders.

1- Run **"1- Data processing & normalization.R"**. Check for results in the "Clean data" folder.

2- Run **"2- Conversion into GIS data.R"**. Check for results in the "GIS Layers" folder.

3- Go to **QGIS** (or any GIS software) and load the **Tunisia_map.shp** found in the "Tunisia_map" folder. This will act as a reference map.

4- In QGIS, go to **Layer > Add Layer > Add Vector Layer > select any shapefile** from the "GIS Layers" folder. The points will be placed on the map.

You can then **click on the layer > Show Labels** to fully display the data.

NOTE: To use the R code, don't forget to change the file paths and make sure all packages are installed.